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Nanjing Tuowei Integrated Circuit Co., Ltd.

NanJing Top Power ASIC Corp.

data sheet

DATASHEET

TP5602 TP5602B Power Bank

Dedicated 4-in-1 single-chip system (low-cost

synchronous 3A lithium battery charging, 5V3A

boost, display, battery protection)

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TP5602, TP5602B mobile power dedicated **4** -in- **1** single- chip system

Overview

TP5602 and TP5602B are only in boost mode, the LED light display mode is different. When TP5602 is working on button boosting or automatic boosting, the LED light will be on for 3 seconds and then off; when TP5602B is working on boosting, it will be on for a long time until the boosting stops and the LED will go off. The other functions, typical application circuit diagrams, precautions, packaging, etc. of the two are the same, and the TP5602 chip is introduced without special instructions below.

TP5602 is a fully integrated function mobile power dedicated VLSI, specially designed for medium and high power mobile power. Internally integrates automatic charging management and battery isolation PMOS2, boost management, power display, battery protection and other functions into a single chip, plus an input isolation PMOS1, a power NMOS1, an inductor, and a small number of RC components, the application circuit is simple , stable and reliable performance, debugging-free mass production, high efficiency and high yield.

TP5602 adopts unique single-inductor multiplexing technology, built-in high-efficiency synchronous step-down 4.2V3A lithium battery charging circuit and synchronous boost 5V 3A output circuit; 4-channel LED display output can display the power during charging and discharging in 5 segments, which can be applied to Red, green and blue LEDs are directly driven, and are also used for abnormal state alarm display; no external battery protection circuit is required, and various input and output protections of the battery are integrated inside (overvoltage, overcharge, undervoltage, overcurrent, short circuit, etc.);

TP5602 provides function settings: plug-in automatic charging, button boost, light load automatic shutdown (when boosting); step-down charging constant current adjustable; 5V boost output constant current adjustable externally. Its ultra-small QFN24 package and simple peripheral circuits make the TP5602 very suitable for other
High-power stand-alone charging or stand-alone boost management applications for portable devices.

TP5602 has a wide input voltage (4.2-7V MAX), and is protected against reverse connection of the power supply. When charging, charging the battery is divided into three stages: trickle precharge, constant current, and constant voltage. The trickle precharge current and constant current charging current are both Adjusted by an external resistor, the maximum charging current is 3.5A. 4LED different display modes when charging different battery voltages. The switching mode of charging switching frequency of 700KHz makes it possible to use smaller peripheral devices, and still maintain a small amount of heat during high current charging.

TP5602 has a built-in 4.25V VIO charging adaptive circuit. When the VIO current is not driven enough to cause the VIO to drop to 4.2V, the internal charging current will be automatically reduced. Users do not have to worry about the problem of low-power USB ports, power supplies or adapters due to excessive current settings. .

TP5602 has built-in anti-battery backflow PMOS2 circuit, so peripheral protection such as anti-backflow Schottky diode is not required.

The TP5602 synchronous boost circuit can output constant voltage and current limit, the voltage is internally fixed at 5V, and the current can reach 3A. The current limit value can be adjusted by an external setting (VIlmt) to limit the output current of the battery terminal, thereby controlling the VIO output power. Soft-start protection during boost startup, output short-circuit and over-current protection (250mS long-term over-current/short-circuit shutdown). External NMOS, cycle-by-cycle current limiting. Typical can drive 5V3A, limit drive 3.5A, 5V3A@Vbat=4V efficiency 93.5%.

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Features

to charge the local battery: ÿ

Single-cell 4.2V lithium battery charging, typical 3A ÿ Wide operating voltage, up to 7V ÿ Bypass output, power adaptive ÿ Anti-power reverse connection, anti-battery overvoltage 4.3V double protection, Safer ÿ Built-in power management PMOS2, fully synchronous switching mode ÿ Trickle, constant current, constant voltage three-stage charging, protection

Battery

ÿ Programmable charging current ISET, 0.1A--3A ÿ Pre-charge trickle: 20%ISET Constant voltage shutdown current: 20%ISET

ÿ 4-way LED charging status indication, the highest digit flashes ÿ Chip over-temperature automatic power reduction protection, under-voltage

Protection ÿ Short circuit

protection at battery terminals ÿ Built-in multiple battery protections, no need for external protection chips ÿ Switching frequency 700KHz, typical inductance 2.2uH ÿ 4.2V±1% charging voltage control accuracy ÿ Using QFN24 4mm*4mm ultra-small package

Pack

Boost circuit: ÿ

Press the button to start VIO, fixed 5V output ÿ Long press the button for 2.5S, the boost is turned off and enters standby model

ÿ Adjustable maximum output constant current, typical output 5V3A ÿ 4-way LED power status indication, the light will automatically turn off after 3 seconds (TP5602B light is long on), the battery is less than 3V and the alarm is 6 times protection, less than 2.4V full shutdown, no action ÿ Chip over-temperature protection, automatically reduce output current, under-voltage, over-current, short-circuit automatic standby protection ÿ Switching frequency 300KHz ÿ Automatic standby after 10 seconds of light load (output <30mA) ÿ Battery current as low as **6uA** after standby ÿ Typical output 5V 3A , 4V Vbat typical output efficiency 93.5%. Maximum output 5V 3.5A.

Absolute Maximum Ratings

ÿ VINÿ -10 ÿ +9V

ÿ VBATÿ -0.7VÿV

ÿ Maximum junction temperature:

145°C ÿ Operating ambient temperature range: -40°C to 85°C

ÿ Storage temperature range: -65°C to 125°C ÿ Pin temperature (soldering time 10 seconds): 260°C

Application

Scope ÿ Power Bank

ÿ Portable Equipment ÿ

Electronic Cigarette

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Typical Application Circuit

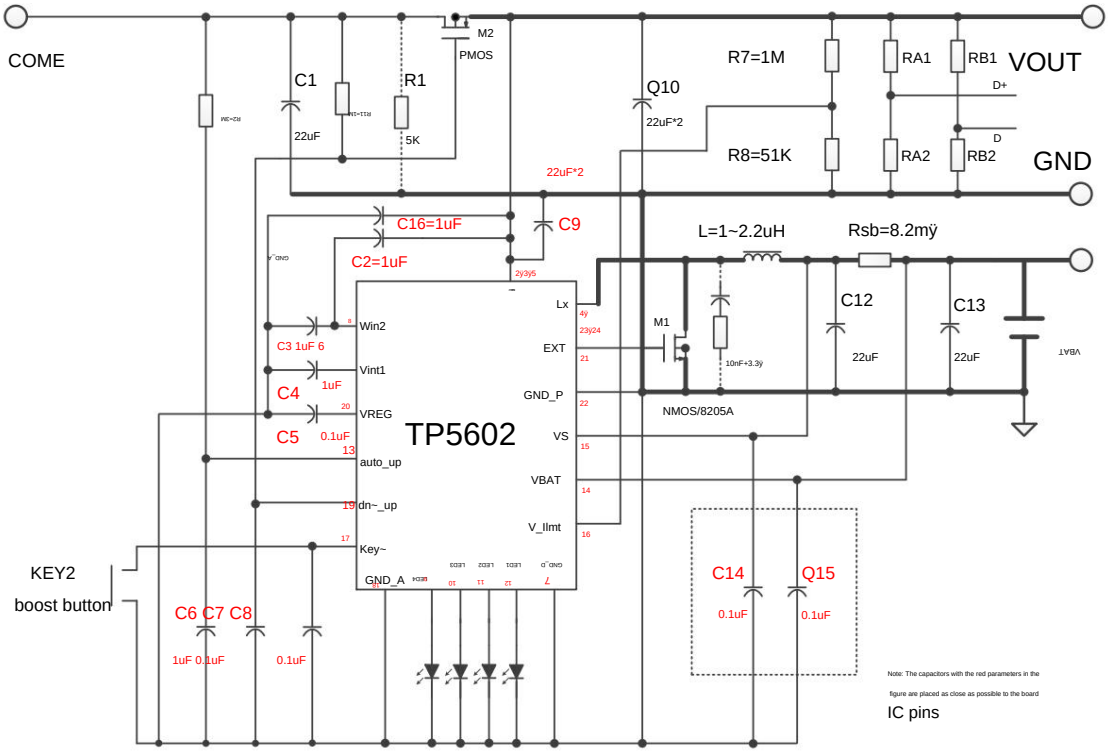
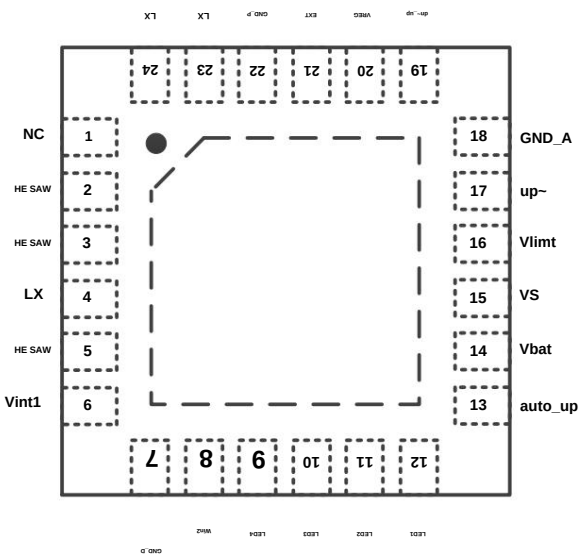


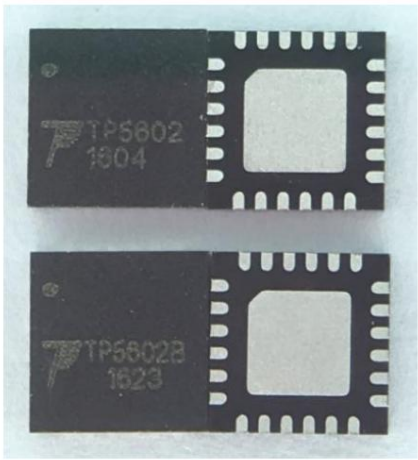
Figure 1 Schematic diagram of TP5602 charging 4.2V lithium-ion battery 3A and boosting 5V3A

Packaging/Ordering Information



Top view of the 24-pin 4mm*4mmQFN24 package (the heat sink ground cannot be connected to other potentials)

Order model
TP5602-QFN24/ TP5602B-QFN24 Device
Marking
TP5602/ TP5602B
Physical picture



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functional block diagram

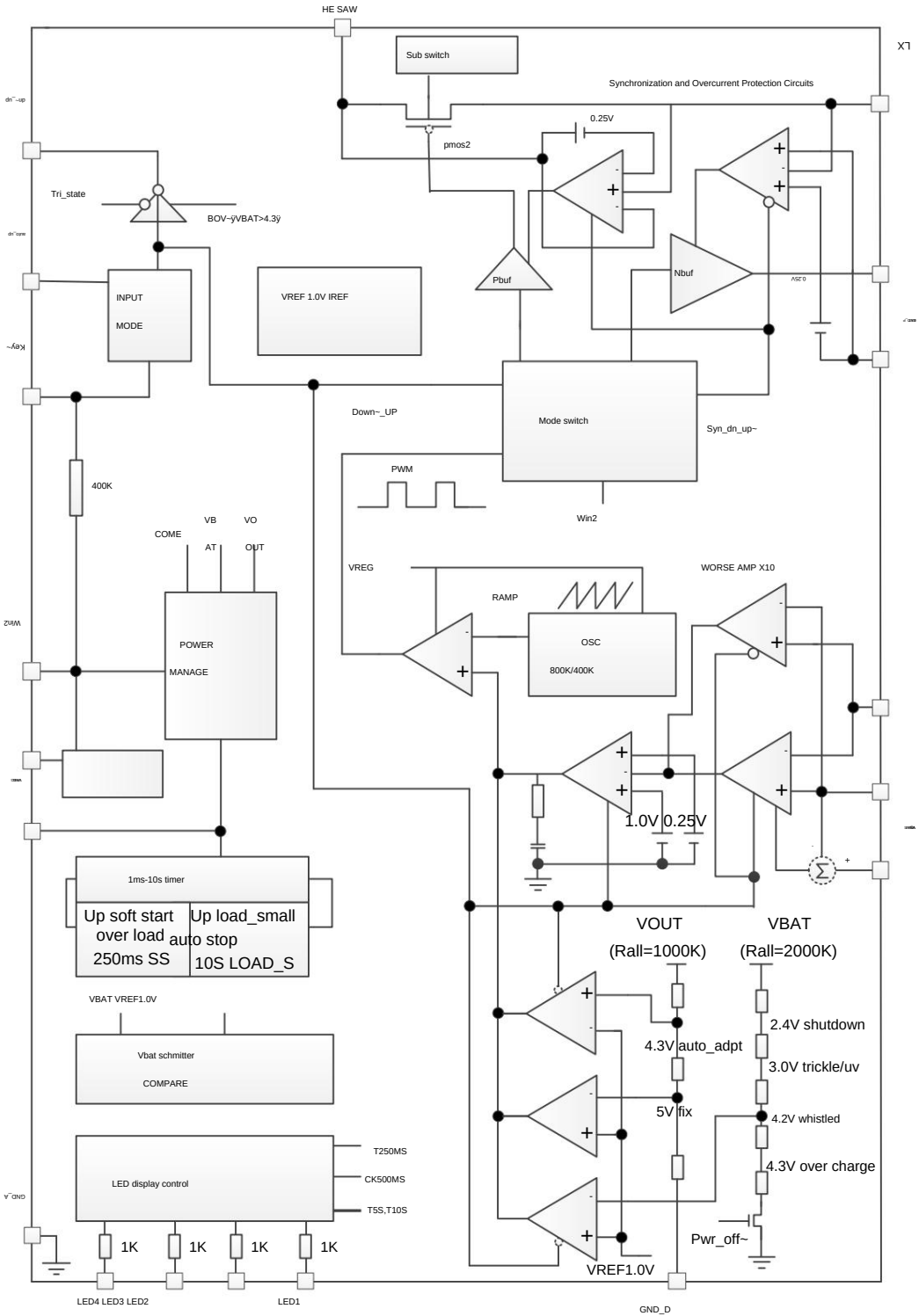


Figure 2 Block diagram of functional modules

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Electrical characteristics

Table 1 TP5602 electrical performance parameters

Where the table note • indicates that the indicator is suitable for the entire operating temperature range, the typical circuit in Figure 1, otherwise it only refers to TA=25ÿ, VIO=5V, unless otherwise noted. Ron_nmos1=15mÿ, Ron_pmos1=60mÿ.

symbol	parameter	condition		Min	Typ	Max	Units
	Step-down charging part						
COME	Input supply voltage		•	4.2	5	7.0	IN
IRVIN	Input supply current (without lights)	Charging mode, RSb=0.1ÿ	•		240		mA
		Standby mode (VB=4.25)	•		550	750	ÿA
		shutdown mode (Vin < VBAT ; or	•		550	750	ÿA
		Vin<VUVÿ			550	750	
VFLOAT	Charge cut-off voltage Rsb=0.1ÿ 4.2V Li-ion battery 4.160				4.2	4.250 V	
Vsb	constant current setting voltage reference accuracy	Rsb=0.1ÿ		22	25	28	mV
IRBAT	BAT pin current:	RS=0.1ÿ, constant current mode	•		250		mA
	(Current mode test conditions are	RS=8.2mÿ, constant current mode	•		3		A
	battery=3.8V)	Standby mode, VBAT=4.25V	•	0	-6	-10	uA
		VIN=0VÿVBAT=4.25V	•	0	-6	-10	uA
IRTRIKL	Trickle Precharge Current	1.2V<VBAT<VTRIKLÿ	•		600		mA
		RS=8.2mÿ					
Fdn	Oscillation frequency			600	700	800	KHz
DMAX	maximum duty cycle				100%		
DMIN	Minimum duty cycle			0%			
VTRIKL	trickle charge threshold voltage (4.2V) RS=0.1ÿ, VBAT rises VTRHYS trickle charge hysteresis			2.8	2.9	3.0	IN
voltage		RS=0.1ÿ		60	80	100	mV
VUV	VIN undervoltage lockout threshold	From VIN low to high	•	3.7	3.8	4.0	IN
VUVHYS	VIN undervoltage lockout hysteresis		•	150	200	300	mV
VADPT	VIN adaptive startup voltage	VIN from high to low		4.15	4.25	4.35	mV
IBTERM	ISET/5 Termination Current Limit	RS=0.1ÿ	•	40	50	70	mA
VLED	LED1-4 pin output driver	ILED=2mA		2.5	3	4.0	IN
ÿVRECHRG	Rechargeable battery threshold voltage	VFLOAT-VRECHRG		100	120	150	mV
TLIM	Chip protection temperature				145		ÿ
RONP2	Power PMOS2 On Resistance				30		mÿ
ILMTp	Power PMOS2 Cycle-by-Cycle Current Limit	0.25V/Ronp2 VBAT			6		A
tRECHARGE	Recharge Comparator Filter Time tTERM	high to low IBAT		300	500	700	mS
Terminate	Comparator Filter Time	drops below CR/5		1	1.8	2.5	mS

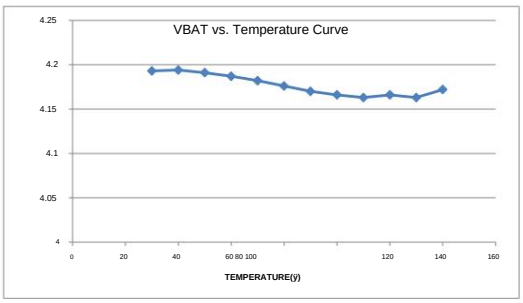
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Unless specified, otherwise, VIN=0, VBAT=3.8V, TEMP=25ÿ, VIO=5V, Rout=5ÿ, Cout=88uF,
Rsb=8.2mÿ, the KEY~ key starts the boost

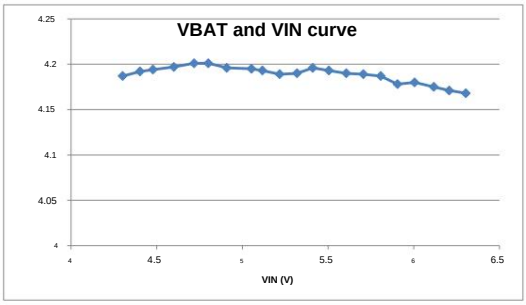
symbol	parameter	condition		Min	Typ	Max	Units
	boost section						
VBAT	Input supply voltage		•	3.0	3.8	4.5	IN
IBAT battery output current (excluding lamps)		Boost VIO=5V, Rout=50K		2	3	10	mA
		Standby mode	•	4	6	15	ÿA
		Shutdown mode (VBAT=2.2)	•	0	0.5	2	ÿA
VOUT	Output Voltage Accuracy	RS=8.2mÿÿIOUT=100mA		4.87	5.02	5.17	IN
VIlmt Constant current setting voltage valid range		VIlmt has VIO which is taken by the voltage divider resistor have to		0		1	IN
IOUT MAX	VOUT (4.8V) maximum current	RS=8.2mÿÿVIlmt=0.1V		2.8	3	3.2	A
Iout_LS	Light Load Shutdown Current Threshold	L=2.2uH (positive correlation with inductance)		20	50	100	mA
Fup	Boost Oscillation Frequency			250	300	400	KHz
DMAX	maximum duty cycle				85%		
DMIN	Minimum duty cycle			0%			
VLMTn	Cycle-by-cycle current limiting of power NMOS1 Voltage	design value		0.22	0.25	0.28	IN
Tdisp	Battery LED display cycle			2.5	3	3.5	S
	TP5602B battery LED display cycle				Long bright		S
Tnoload	Light load detection period	IOUT=200mA down to 5mA		5	10	20	S
VuvB3	Battery undervoltage standby point	VBAT from high to low		2.8	3	3.2	IN
VuvB2	Battery too low shutdown point	VBAT from high to low		2.1	2.4	2.6	IN
VLED	LED1-4 pin output driver	ILED=2mA		2.5	3	4.0	IN
TO THE TEMPLE	Chip protection temperature				145		ÿ
Tss_up	Boost soft-start time	VIO=0 to VBAT(COUT=100uF)		100	500	2000	uS
Battery current when	IBAT_UP_ST output is shorted			50	100	200	mA
TUP_ST	VIO short circuit detection time	VIO=0V		200	450	600	mS

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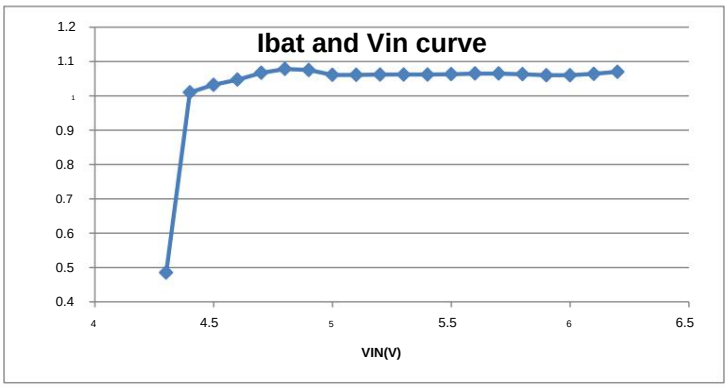
Typical performance index (Lithium battery charging mode)



The relationship between cut-off voltage and ambient temperature

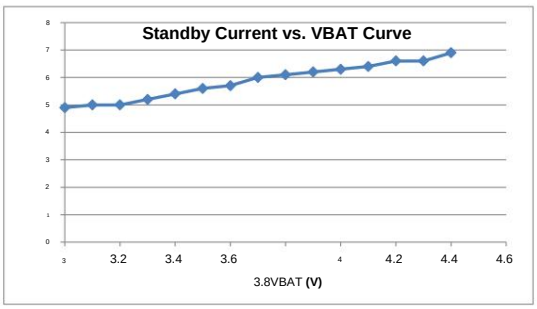


The relationship between cut-off voltage and power supply voltage

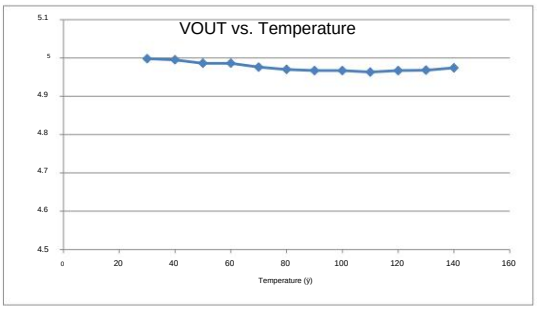


The relationship between charging current and supply voltage

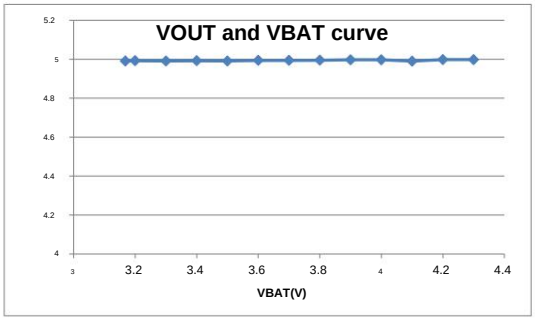
Typical Performance Specifications (Boost Mode)



Standby Current vs. Battery Voltage

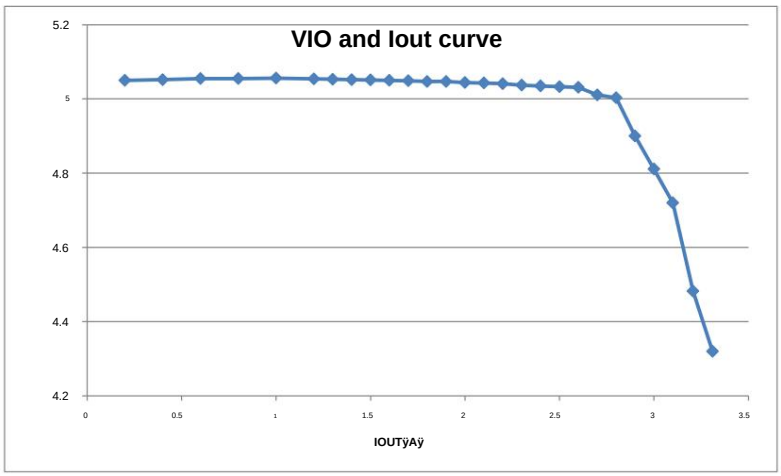


VOUT vs. Ambient Temperature

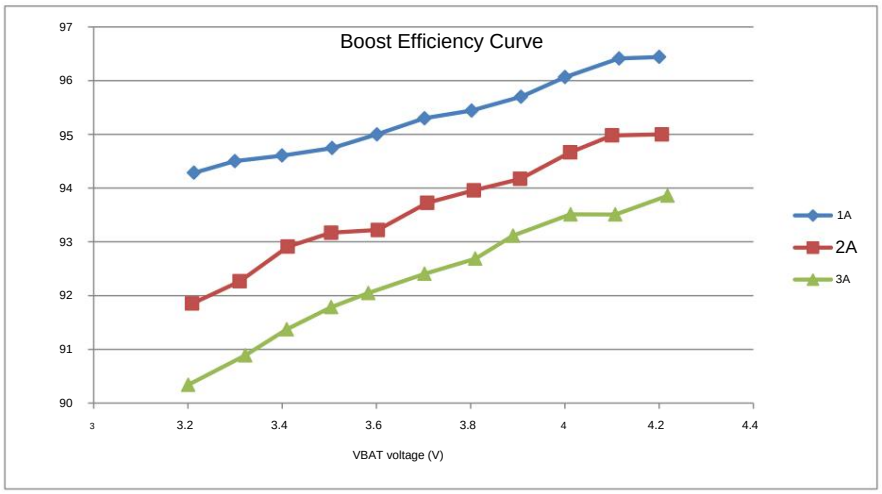


VOUT vs. Battery Voltage

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VOUT vs. OUTPUT CURRENT



Boost Efficiency vs. VBAT

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Pin Function and Instructions for Use

VIO (pins 2, 3, 5): 5V input/output common port is connected to an external 5V power supply when charging, as the local battery charging power supply, the system automatically switches to charging mode (if it was originally in boost mode, it may be delayed for 6S -12S until the boost is automatically stopped); when boosting (key~ is pressed), it is used as a 5V output port.

VIO is also the main power supply of the internal working power supply, so the position of the capacitor is very important to the output accuracy, and it needs to be arranged first and close to the chip port. See the PCB board precautions for details. For 5V2A, it is recommended to connect an external ceramic capacitor of 2*22uF. If the VIO ripple is large or unstable under high current, it is recommended to connect a larger capacitor. For example, for 5V3A, you need to connect 4*22uF or 2*47uF, or larger and more stable. **Such as use when externally connected to PMOS1 :** When the chip is in standby mode (charging mode: dn~_up=0), VIN is output to VIO through PMOS1 tube. When VIO>4.25V, TP5602 automatically enters charging mode. When the chip is in boost mode (dn~_up=VIO=5V), the protection PMOS1 tube is turned off, VIN does not output to VIO, and VIN will not automatically trigger the charging mode when it is powered on. Only when the boost mode ends, dn~_up When = 0, the charging mode will be entered; but if the auto_up pin is connected to VIN through a resistor, the VIN can be automatically detected to be powered on. When auto_up>3.5V, the boost will be automatically turned off and the charging mode will be activated.

GND_A, GND_D (pin 18,7) internal ground

GND_P (Pin **22)** drive and power ground

LX (pins **4, 23, 24): Built-in power PMOS2** tube, external **NMOS** drain connection point.

LX is connected with the external inductor as a switch terminal for battery charging and discharging, and also as a synchronous voltage and over-voltage. The detection terminal of the current voltage.

Vint1 (Pin 6): Internal Power Supply 1. External capacitor **1uF.** When working (charging or boosting) start = VIO, when standby = 0;

Vint2 (pin 8): Internal main power supply 2, external capacitor 1uF. Always output the highest power of the system.

It is automatically switched to VBAT by the internal power switching circuit, and the highest level in VIO is used as the internal working power supply. When working = VIO, when standby = VBAT. In order to prevent the transient of VIO from being too large when switching modes, the transients of the two power supplies of Vint2 and VIO are different, and internal misoperation occurs. A 1uF capacitor needs to be connected across Vint2 and VIO to suppress power transient differences.

LED4-1 (pins 9, 10, 11, 12): The positive pole of the external light-emitting diode **LED ,** the maximum level of normal operation is 5V, and it can drive blue light, white light, red, green **LED ,** etc.

Normal working power indicator port; abnormal state alarm indicator. There is a 1K ohm current limiting resistor inside. Users can also choose to add series resistance according to different LEDs and brightness, see LED instructions for details.

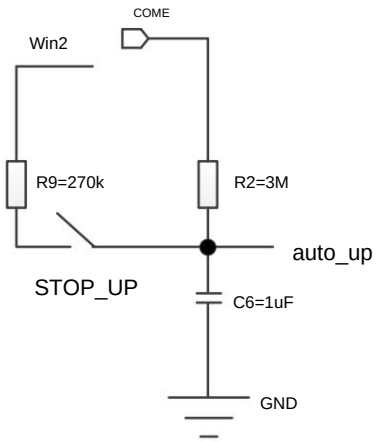
auto_up (Pin 13): Mode Detect Port 2. The typical usage is directly connected to **GND;** when it is high, it is forced to charge mode (boost is prohibited); the falling edge is forced to start a boost; when it is always low, the mode is determined by the KEY~ key; (in charging or standby mode, charging or not depends on whether **VIN** or **VIO** has voltage). Usage 1: The 3M resistor is connected to the VIN terminal (in other applications with external PMOS1, see Figure 3 on the next page), after detecting a high level, it will automatically stop the boost mode and enter the charging mode;) automatically start boost mode, VIO output

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5V, boost, automatic shutdown at light load. The connection method is as shown in Figure 3:

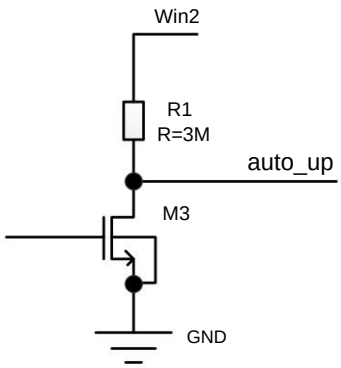
(VIO input/output single port scheme Figure 1, usage 1 is invalid)

Usage 2: After the switch is pressed, the port is connected to the high level of Vint2, the boost mode is manually turned off, and the boost mode is started after release, and the light load automatically stops. The connection method is as follows:



Usage 3:

The external MPU controls the charging mode and boost mode, the NMOS open-drain end of the microcontroller is directly connected to auto_up, and the Vint2 high level is pulled up through the resistor. MPU closes (pulls the port high) or opens (gives a falling edge) boost mode (no KEY~ button), and automatically shuts down under light load. The connection method is as follows:



If it is directly connected to 0 potential, it will not automatically switch to boost mode when VIN is powered down. VIO output is 0. To prevent false VIN glitches during mode switching Triggered, the auto_up port needs to use the RC delay loop, the larger the COUT, the larger the RC delay required. Generally >500mS, (3M, 1uF).

VBAT (Pin 14): Battery Voltage Terminal. Connect the positive terminal of the battery to this pin.

VS (Pin 15): Battery input/output current detection terminal.

Rsb is connected to VBAT and VS as a sampling resistor for battery current, and there is an automatic current polarity identification circuit inside.

Rsb sets the maximum charging current value, V_Ilmt sets the maximum discharging current value, and other currents are automatically adjusted internally.

Vsb=VS-VBAT

During constant current charging Vsb=25mV Maximum input current ISET0=25mV/Rsb During trickle charging Vsb=5mV Maximum input current ISET1=5mV/ Rsb During boost discharge Vbsmax=4*25mV The maximum discharge current limit is about ISET2=4*ISET0(When there is no voltage modulation, VIO=5V, V_Ilmt=1/5VIO)

ISET2 is determined by the internal V_Ilmt (VIO sampling voltage). During modulation: ISET2 decreases when V_Ilmt decreases, V_Ilmt=[0-1]; ISET2 is jointly determined by Rsb and V_Ilmt.

E.g:

Rsb=25mÿ, V_Ilmt=0.5V(=1/10 VIO) , then charging constant current ISET0=0.025/0.025=1A

5V OUT

ISET2=(0.025+0.075*0.5)/0.025=2.5A Output current limit IOUTÿ0.9*4.2*2.5/5=1.8A

If **V_Ilmt=0.25(=1/20 VIO) then** ISET2=(0.025+0.075*0.25)/ 0.025=1.7A output current limit IOUTÿ0.9*4.2*1.75/5=1.3A

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Battery current setting table (customers must take the actual measurement as the standard, the following

The table is the test value of our demo version)

Rs milli Europe	Charge Limiting (A)	V_Ilmt= 5* R8/(R7+R8)	Iout(A) (vout=4.8@ VB=4.2)	Iout(A) (vout=4.8@ VB=3.6)
25 1.0		0.5	1.9A	1.76A
25 1.0		0.12	1.0A	0.90A
25 1.0		0	0.8A	0.77A
12 2		0.5	3.5A	3.40A
12 2		0.1	2.0A	1.90A
12 2		0	1.6A	1.49A
8	3	0.5		
8	3	0.1	3.0A	2.94A
8	3	0	2.5A	2.34A

V-Ilmt (Pin 16): Battery Discharge Current Setting end.

When charging, the port does not work, when discharging Set different voltage initial value 0-1V, (internal 1V clamp circuit). Can change the battery discharge maximum Current limit multiple. If sampling through a voltage divider resistor VIO, the input limit modulated with VIO can be obtained flow value (in conjunction with Rsb), thus keeping the output The current is constant.

key~ (Pin 17): Boost Mode Port **1**

External key, low pulse is valid. After pressing the button Start boost mode (when auto_up=low):

Such as VBATÿ3V; or VIO short-circuit overcurrent wait, the LED abnormal alarm flashes 6 times and then automatically waits for machine, no boost.

If VBAT>3V, the battery indicator will display about Turns off after 5 seconds. VIO outputs 5V power supply. again Press key~ to display the battery again for 5 seconds. If even Continuous output light load, (such as Ioutÿ30mA), about 10 Automatically stop after seconds, turn to standby mode, battery Power consumptionÿ6uA.

Long press the button for 2.5S boost to stop.

dn~_up (pin 19): Multipurpose protection Gate input of **PMOS1 .** Also working status indicator mouth:

dn~_up=0 charging mode, synchronous step-down mode, the external PMOS1 is turned on (scheme 3), PMOS2 is the charging switch tube, NMOS is the step-down Sync tube.

When the system is damaged, such as PMOS2 punch-through, etc. VBATÿ4.3V overvoltage, the protection circuit works:

dn~_up=high resistance state, by adding 1M resistance Pull to VIN protection tube PMOS1 to force close, Stop charging.

External C7=0.1uF capacitor to prevent When VBAT=0V, the possible charging does not start

dn~_up=VIO(5V) boost mode, PMOS1 is turned off, NMOS is the discharge switch, PMOS2 is a boost synchronous transistor.

VREG (Pin 20): Internal Power Supply 3.

VREG is an internal secondary power supply that Connect a 0.1uF bypass capacitor to ground.

VREGÿ4.0V. only when working active, 0V in standby.

EXT (pin 21): External **NMOS** gate driver end.

Expand a power NMOS, switch signal, CMOS output, high level is VIO, low level is 0. Can drive NMOS up to Cg=2000p power tube.

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Working principle description:

TP5602 is specially designed for the need of built-in single section 4.2V lithium-ion battery charging, while requiring high-efficiency high-power 5V stable DC power system designed. Since the chip integrates a variety of battery input and output protection circuits, there is no need for a battery protection board or circuit outside. Plus built-in 4-way power display and alarm display. The unique multiplexing technology of inductors and power tubes uses a set of devices and automatic switching modes of boost and buck to complete high-efficiency and high-current synchronous buck charging and synchronous boost output. The production complexity of the system is greatly reduced, and the system reliability and safety are greatly improved while the cost is reduced. External

PMOS1: The unique design of multi-purpose protection and bypass PMOS1 can not only ensure the side charging and bypass output at the same time, but also prevent the reverse connection of the charging power supply. When the battery voltage is over-voltage, PMOS1 will also be automatically cut off, and the charging voltage limiting circuit of PMOS2 is added to double protect the battery. PMOS1 is turned off during boost to prevent VIO from leaking back to VIN.

Charging

process: In standby mode, PMOS1 is on, dn~_up=0, VIO=VIN.

After VIN is powered on: the charging circuit is not activated when VIO<3.8V. The battery has no current input; the LED light is off; when 3.8<VIO<4.3, the charging circuit is not activated, and the LED light displays the power; when VIOÿ4.3V, the charging circuit

is activated, and the synchronous step-down mode is activated, PMOS2 is inductive charging, and NMOS is synchronous tube, the current flows through VS external Rsb to the battery. Vsb is the sampling voltage, which controls the duty cycle of the switch and the charging current value.

The LED light shows the

power as follows;		
0- 3.0 LED1/	No long on for 0.5 sec	50% flashing 3.0-3.5 LED1/
No 1 sec 25% flash 3.5-3.8 LED2/LED1	flash 0.5 sec 50%	
LED3/LED2-1	1 sec 25% flash 4.0 -4.2 LED4/	
LED3-1 25% flash for 1 second		
4.2	No flicker/LED4-1	

The charging process

is divided into: Trickle (0<=VBAT<3.0): Vsb=5mVÿ

LED1 flashes at high frequency, other LEDs are off.

Constant current (3.0V<VBAT<4.18): Vsb=25mV The highest digit of the LED flashes slowly, and the other low digit is long on. Constant voltage (4.18<VBAT<4.20), all four LEDs are always on, not flashing, and fully charged. Constant current setting: ISET0=Vsb/Rsb=25mV/Rsb Trickle current: ISET1=5mV/Rsb

During the constant voltage stage, the current will drop sharply with the increase of VBAT, until the charging current is about 20% of ISET0 and below ISET0 for more than 1.8mS, generating a full signal: The LED is always on, and the step-down tube is turned off. Automatic recharging: the battery is fully charged to 4.2V and then standby, if the battery voltage drops to about 4.08V, restart the charging process until VBAT returns to 4.2V. When the internal over-temperature reaches **140** degrees, the protection circuit automatically reduces the output current until the internal temperature no longer rises. This feature allows users to not worry about chip overheating and damage.

Function selection after **VIN**

is powered off: When VIN is powered down to 4.3V, the adaptive circuit starts to automatically reduce the output current until VIN no longer decreases.

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The battery is used as the power source, and the power reset or restart is avoided.

VIN power off:

auto_up port, not fixed to VIN, then auto_up and key~ can be used as manual mode switching buttons. After VIN is powered off, there is no automatic conversion to boost function, and it is directly in standby; however, the key~ key is not prohibited during charging (stop charging mode, enter boost mode, VIO=VIN and boost 5V, whichever is higher). After the auto_up key, it will enter charging or standby mode (stop boost mode). The auto_up port RC loop is connected to VIN (such as the typical application

circuit), then VIN is in automatic charging mode after power-on, no matter what the previous state, the key~ key is invalid. After VIN is powered off, it will automatically switch to boost mode (without the key~ key), and the key~ key is also valid. It is recommended to use this feature option.

Discharge boost process:

In a typical application, both power-down and key~ low-level pulses can trigger the boosting process of VIO from 0V to 5V. At the same time, the LED will turn off after about 3 seconds to display the **power** (TP5602B is always on). When boosting,

PMOS1 is turned off, dn-VREG40, VIN=0V, EXT drives NMOS switch inductor, PMOS2 is boost synchronous, and VIO is pumped to 5V DC for charging portable devices such as mobile phones and tablets ,use. Internal VIO adaptive soft-start current limiting

(VIO rises, the output current is allowed to be larger), cycle-by-cycle current limiting, limiting average battery current, chip over-temperature protection, long-term (250mS) short-circuit and over-current shutdown protection and other multiple protection functions, external battery protection is not required circuit. It also reduces the loss of the external protection circuit and improves the battery power conversion efficiency.

When the **TP5602** boosts the voltage, the LED lights are shown as follows: Battery voltage flashes (discharge mode) 0 ~ 2 times, light off

2.4- 3.0 LED4-1/no long and fast flashing 6 times under voltage alarm		
3.0-3.5 None/LED1	None/LED2-1	
3.5-3.7	None/LED3-1	
3.7-3.9	None/LED4-1	
3.9-4.2		
VIO overload LED4-1/no fast flash 6 times Chip over		
temperature LED4-1/no fast flash 6 times		

When the **TP5602B** is boosting, the LED lights are shown as follows: Battery voltage flashing (discharge mode) 0 ~ 2 times, light shutdown, fast flashing 6 times

2.4- 3.0 LED4-1/No long light under voltage alarm		
3.0-3.5 None/LED1	None/LED2-1	
3.5-3.7	None/LED3-1	
3.7-3.9	None/LED4-1	
3.9-4.2		
VIO overload LED4-1/no fast flash 6 times Chip over		
temperature LED4-1/no fast flash 6 times		

Automatic stop during boost operation:

After the boost circuit starts to work normally, if the output has a load and the load is large enough, and IOUT>50mA under typical conditions , the booster circuit will always work without stopping; if the light load IOUT<50mA (according to Temperature, battery voltage, inductance value, etc. will be different), generally indicates that the external charging is over, or no load is connected, after continuous detection for about 10S, the chip will automatically stop boosting and turn to standby mode;

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In standby mode, the voltage can be boosted again (triggered by low-level pulse of key~ key), or it can automatically enter **charging** (VIN >4.3V). In standby mode, VIO, VIN and **VBAT** are completely isolated, and the power consumption of the chip system is only **6uA.**

Manual or external control stop required when boosting

Regardless of light load or not, the user can pull up the auto_up port to a high level Vint2 through a switch or resistor. The boost stops immediately, and the key~ key does not work. At this time, it is in standby state, and you can enter the charging mode; after auto_up becomes low, the chip automatically returns to the boost state (regardless of whether the key~ key is pressed).

The connection method of CPU control

shutdown: CPU open-drain NMOS port output pull-up 3M Ohm resistor connected between auto_up and Vint2. The large resistance 3M resistance is to reduce the battery standby current.

Key~ pin control shutdown method: In a typical circuit, press and hold KEY2 for 2.5S to realize boost shutdown.

VIO and output current: There is a sampling resistor with VIO=5V fixed output inside the chip, and the user does not need to adjust the resistor externally. The maximum output current is determined by the value of Rsb and V_Ilmt, (see VS port settings). The maximum charging current is set by Rsb and has nothing to do with V_Ilmt;

The maximum discharge current of the battery is set externally by V_Ilmt to 1-4 times (corresponding to 0V-1V V_Ilmt) of the maximum charging current. The built-in VBAT voltage and VIO modulation and boost efficiency (typically 93.5%) together determine the maximum value of IOUT during boost. VIO=5V falls to VBAT, IOUT rises.

VIO remains unchanged, VBAT decreases, and IOUT decreases.

If VIO is less than VBAT continuously for 250mS, it means the output is overloaded or short-circuited, the internal automatic shutdown protection, and **6** flashing lights alarm overload.

Battery under-voltage

protection: When boosting, if the battery voltage is less than 3V, the Automatic shutdown, and flashes **6** times to alarm undervoltage.

If the battery voltage is less than 2.4V, in order to prevent the lithium battery from being damaged, all the chips are shut down and locked. When the battery current drops below 1uA, the external keys are invalid. In the future, even if the battery voltage returns to above 2.4V, the lock will not be unlocked. Only after VIN is increased to a high level, it will automatically reset after charging once, and then the lock can be unlocked. (Note: When the engineer is testing the chip or production, if the VBAT is shifted for a short time, it may happen that the capacitor voltage of CBAT is less than 2.4V, and the VBAT will not work when it is powered on again . The capacitor is discharged to 0 and then VBAT power-on reset is also available).

Chip Internal Thermal Limiting

Whether in charge or discharge mode, an internal thermal feedback loop will reduce the set battery current if the chip temperature attempts to rise above a preset value of approximately 145°C. This feature prevents the TP5602 from overheating and allows the user to increase the upper limit of the power handling capability of a given board without risking damage to the TP5602. The charge/discharge current can be set based on typical (rather than worst-case) ambient temperature, provided the SoC guarantees that it will automatically reduce the current under worst-case conditions.

Various types of ceramic capacitors can be used for input VIN, output **VIO, VS, and VBAT** terminal capacitors, but high-quality power capacitors are required. It is best to add another 0.1uF ceramic capacitor for bypass, and the connection position must be close to the chip pin. Other port capacitors can use non-power small ceramic capacitors.

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In particular, the capacitor of VIO can increase the capacitance and power capacity as much as possible (for example, when outputting 5V2A, use two 22uF or 47uF ceramic power capacitors in parallel with two VIO pins. If the VIO ripple is large or unstable under high current, the It is recommended to connect a larger capacitor. For example, when setting the output 5V3A, the VIO terminal needs to be connected to 4 1206 package 22uF or 2 47uF, or larger and more stable), and it needs to be very close to the chip VIO pin.

Thermal

Considerations Despite the small size of the QFN24 package, its heat dissipation characteristics are good, however, it requires design coordination of the PCB layout, preferably a thermally well-designed PCB layout to maximize the available charge/discharge current. The thermal paths used to dissipate the heat generated by the IC run from the die to the leadframe and through the die bottom heat sink and vias to the PCB backplane copper surface. The copper area where the pins are connected should be as wide as possible and extend out to the larger copper area to dissipate the heat to the surrounding environment. Additional vias to inner or back copper circuit layers are recommended to improve the overall thermal performance of the charger. Other heat sources on the board that are not related to the charger must also be considered when designing the PCB layout, as they will have an effect on the overall temperature rise and maximum charging current.

The inductor

selects FOSC=700KHz when charging, and reduces the loss of MOS tube when discharging. The frequency reduction of FOSC is 300KHzÿ

According to different output current requirements, the inductance value is 1uH-2.2uH, 2.2uH is recommended for boost output 5V2A, and 1uH is recommended for output 5V3A.

The rated current of the inductor is recommended to use a power inductor with a smaller internal resistance than twice the boost output current. Under the same circumstances, if a higher inductance value is selected, the system efficiency will be higher.

See the following pages for other application solutions and precautions of **TP5602 :**

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TP5602 other application circuits and devices

Figure 3 and Figure 4 are other application circuit diagrams of TP5602:

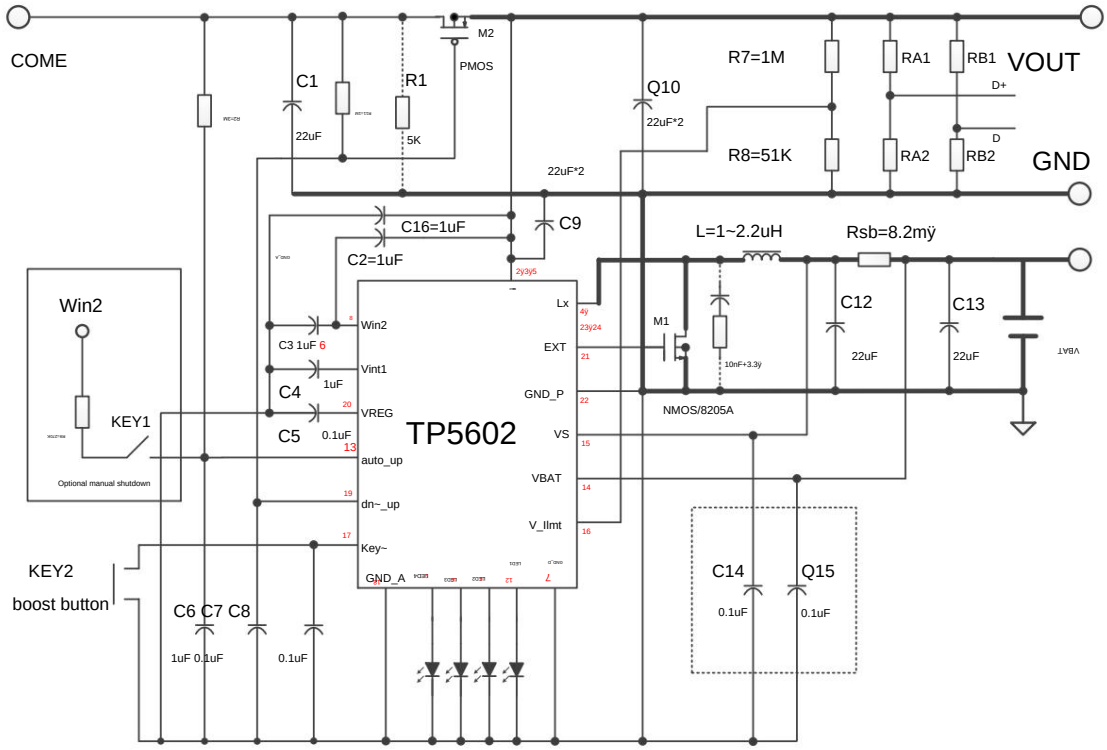


Figure 3: Manual or CPU Controlled Shutdown at Boost

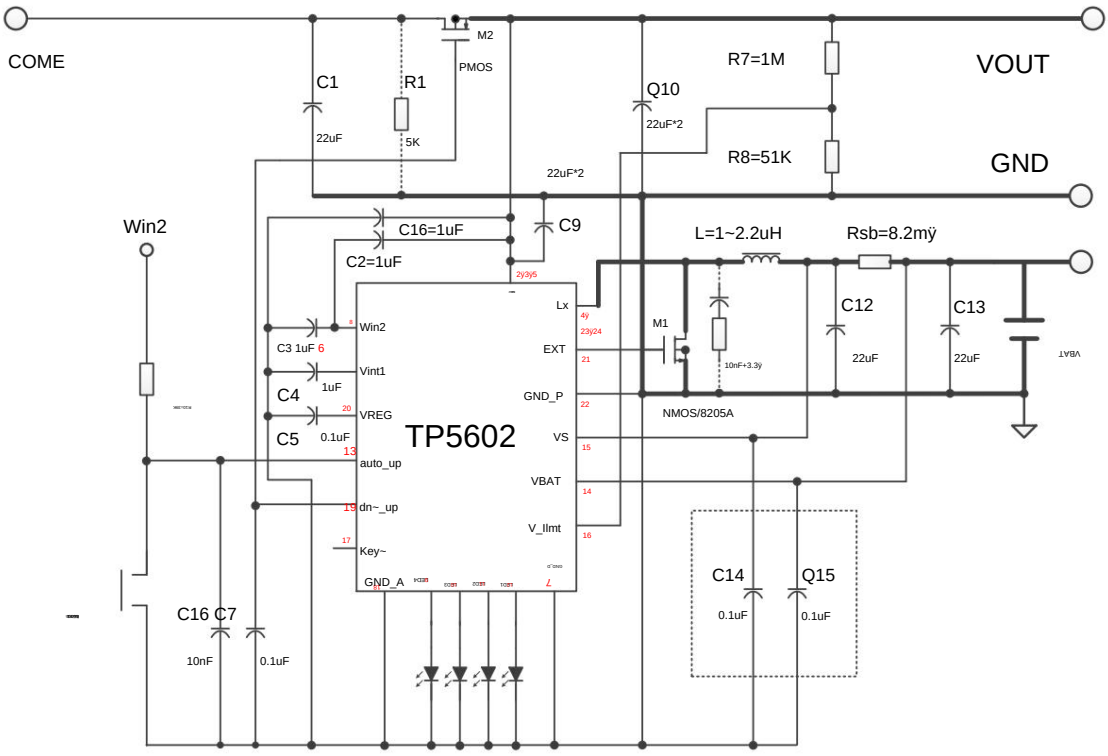


Figure 4: 15W push-button electronic cigarette solution

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The following table is a recommended list of components for a typical application circuit:

Table 2 4.2V Li-ion battery 3A charging, 5V 3A boost

No.	Component Name	Model & Specification	Tag usage		Remark
1	Main control IC	TP5602 QFN24 -4*4	U1	1	
2	SMD power inductors	double winding 1.0uH/10*10	L	1	Saturation Isat, temperature rise current **Idc** Greater than **7A, DCR** less than **0.01, sense value 1.0uH**
3	SMD capacitor 0.1uF/0603 10%		**C5 C7 C8 C14 C15**	5	The withstand voltage is not less than 16V, it is recommended Use chip ceramic capacitors, red The color number capacitor is in the circuit should be as close as possible to the chip pins place
4	SMD capacitor 22uF/1206 10%		Q1 **Q9** Q10 C12 C13	7	
5	SMD capacitor 1uF/0603 10%		**C2 C3 C4 C6 C16**	5	
6	SMD capacitor 0.01uF/0603 10% 7 SMD resistor 3.3R/		160	1	**EMC** test ready device
0805 1%			RLX	1	
8	SMD resistor 0.0082R/1206 1%		RSB	1	Accuracy 1%, package is **1206** or above
9	SMD resistor 5.1K/0603 5% 10 SMD resistor 3M/0603		R1	1	
5% 11	SMD resistor 1M/0603 5% 12 SMD resistor 51K/		R2	1	
0603 5% 13	SMD LED 0603 Blue LED1~4 14 SMD MOS		R7 R11	2	
8205A/TSSOP8 15	SMD MOS 2301/SOT23-3 16 Key		R8	1	
switch 6.5mm*5.1mm	Micro USB female socket			4	For battery indicator
			M1	1	Synchronous **NMOS**
			M2	2	switch **PMOS**
			KEY2	1	**Button + socket**
17	Mini USB	5-pin full stick	1 for input		
18	USB female	USB-A female socket sticker Sheet/white, 4P USB Female socket type A	1 for output		

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Precautions for using **TP5602**

1. The capacitors in the circuit should be as close as possible to the chip, not the port of the PCB board. Take the VIO terminal capacitor as an example: When the output is 5V2A, the VIO terminal needs to be connected to two 22uF capacitors, one on the VIO pin side of the chip, one on the USB output port, or both on the VIO pin side; when When the output is 5V3A, the VIO terminal needs to be connected to four 22uF capacitors. At this time, two 22uF can be connected to the VIO pin of the chip, and two 22uF can be connected to the USB port.

2. Use high-quality ceramic capacitors for VIO, VS, VBAT, and VIN. VIO pin is recommended to use 1206 package installed ceramic capacitors.

3. Please select a power inductor with sufficient current capacity for the inductor, mainly considering the current requirements during boosting, generally large 2 times the maximum current of IOUT. The inductance value is between 1uH~2.2uH. 4. The traces for VIO and LX through the current loop should be wider than other current signal traces. 5. Pay attention to the node position of each capacitor ground wire. The port capacitors of Vint1, Vint2, and VREG should be connected to the same side ground wire as much as possible, preferably close to GND_A, GND_D, and GND_P. VIO, VBAT, and VS use the same side ground wire, which is well connected to the PCB VIO ground. 6. When using the chip in high current operation, the good connection between the heat sink at the bottom of the chip and the back PCB should be considered.

Ensure good heat dissipation.

7. TP5602 QFN chip maintenance, disassembly and assembly method: TP5602 is an ultra-small QFN24 package. It is recommended to use a heat gun with adjustable temperature when disassembling and assembling. Dismantling: Define the temperature of the air outlet of the hot air gun as 280-300 degrees. The air outlet is first aligned with the four sides of the QFN to preheat for 20 seconds. The air outlet is within about 10mm of the center of the chip, and then the center of the chip is continuously blown to heat the chip, about 20~60 Seconds (different time for different heat sinks), when the chip can be moved, quickly remove it with tweezers. If there is a large area of vias or heat sinks on the back of the chip, it is recommended to use a soldering iron and a solder gun to suck most of the solder from the back hole.

Installation: 1) Use a soldering iron to evenly fill a small amount of tin at the PCB QFN patch pins, or use a soldering iron to fill the PCB QFN pins Apply a thin layer of solder paste directly to the feet. Another chip pin can also be filled with a small amount of tin. 2) Pick up the QFN chip with tweezers, and blow the heat gun at the bottom of the chip (with feet) for 5 seconds. Quickly apply a layer of flux to the bottom surface.

3) Use a heat gun to preheat the QFN pads on the PCB, and place the QFN chips in the correct order according to the pins. The sequence is placed in the center of the soldering position of the PCB, and it can be roughly aligned.

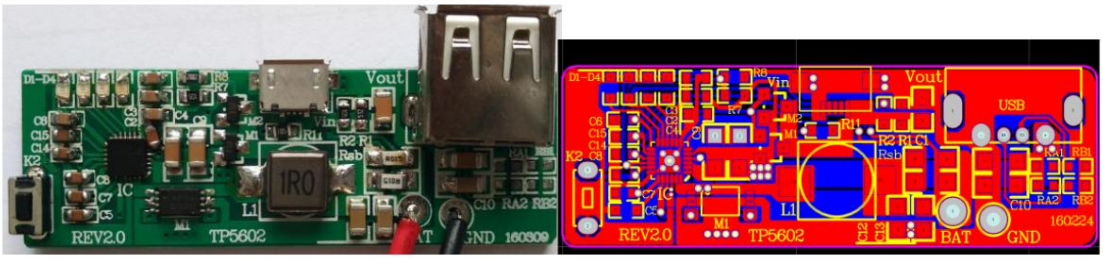
4) Use a hot air gun to blow the chip from just above the chip for 20~60 seconds. If necessary, heat can also be blown around the chip, and the chip will automatically align the pin position due to the surface tension of the melted solder. If the position difference is too large, you can use tweezers to adjust slightly while blowing, and use it after cooling.

For other more precautions, please refer to: TP56 Series Application Notes.

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TP5602 demo board circuit and instructions

The following figure is the demo graphic and PCB schematic diagram of TP5602 (common to TP5602B), using typical application Figure 1.



The demo version and the schematic diagram of the PCB use the typical application in Figure 1.

Note: The back of the PCB board is not provided, it is a large area of ground wire and vias, and there are no components. In order to improve the efficiency, the copper foil of the demo version is thickened to 2OZ.

TP5602 demo version pin and port description: charge 4.2V 3A, discharge 5V 3A Input: Micro USB. Output: USB.

BAT: battery terminal.

Connect the power supply and battery correctly, then it can be charged. The LED flashes step by step during charging. When charging is complete, all four lights are on. For details, please refer to the charging LED display. The K2 key is a boost button. When not charging, press it to start boosting.

K1 is the boost shutdown switch. Press the boost to stop, release to start a boost. In the constant current charging

stage, an ammeter can be connected in series with the positive terminal of the battery to detect the charging current, and the ammeter uses a large range. Considering the internal resistance of the ammeter, it is not recommended to connect the ammeter in series in the test final charged battery voltage mode. The internal resistance will cause the actual voltage of the battery after shutdown to be lower than the shutdown voltage of the chip itself.

Demo layout special notes:

GND: The layout of the ground wire is very important. The chip has digital ground, analog ground and power ground. It is recommended to connect the three ground wires together with the chip heat sink. The output ground wire is mainly the Vout port. In the schematic diagram, the ground wire has been distinguished by thick and thin lines. The thin line is the static ground, and the bold ground wire is the power ground. You can refer to it during layout. It is recommended that the ground wire of the small capacitor around the chip be connected to the static ground.

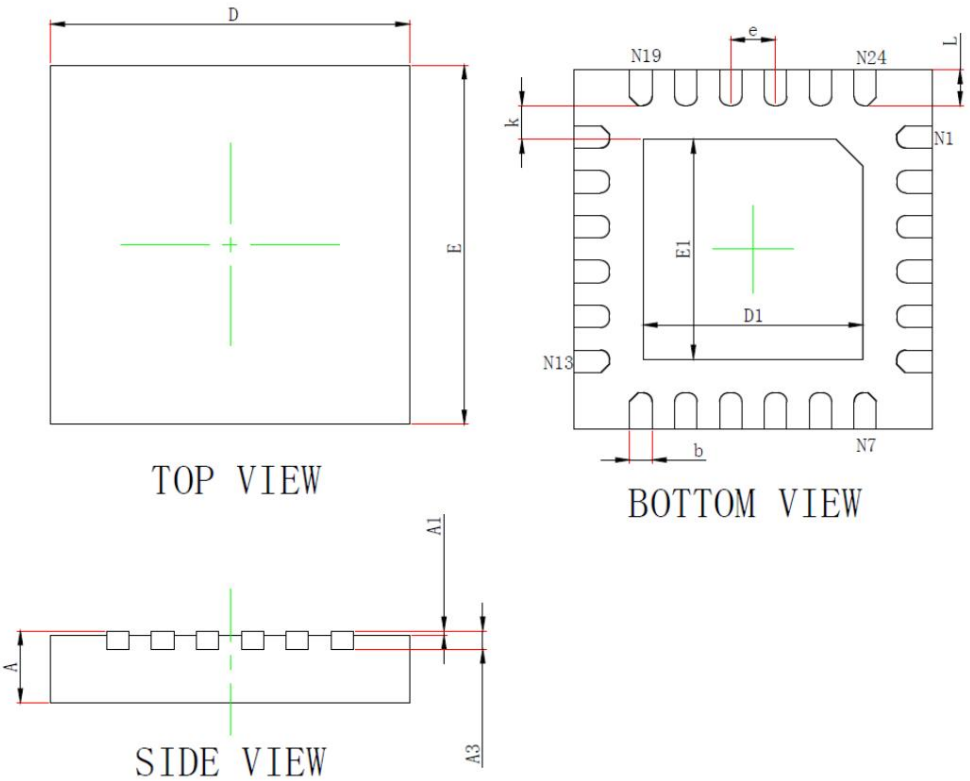
VIO terminal capacitor: VIO has three PINs, which need to be connected together, and need to connect 2~4 22uF capacitors (according to the actual application current, the larger the current is, the larger the external COUT capacitor needs to be, such as 2*47uF capacitor, larger and more stable), and close to the chip port. The capacitor ground should also be connected to the high current ground.

Note: It is recommended that the QFN24 library file pin length in the PCB is slightly larger than the chip length (our company recommends that the PCB layout pins are 1mm long and 0.28mm wide) for ease of soldering.

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Package description

4mm*4mm 24-pin QFN package

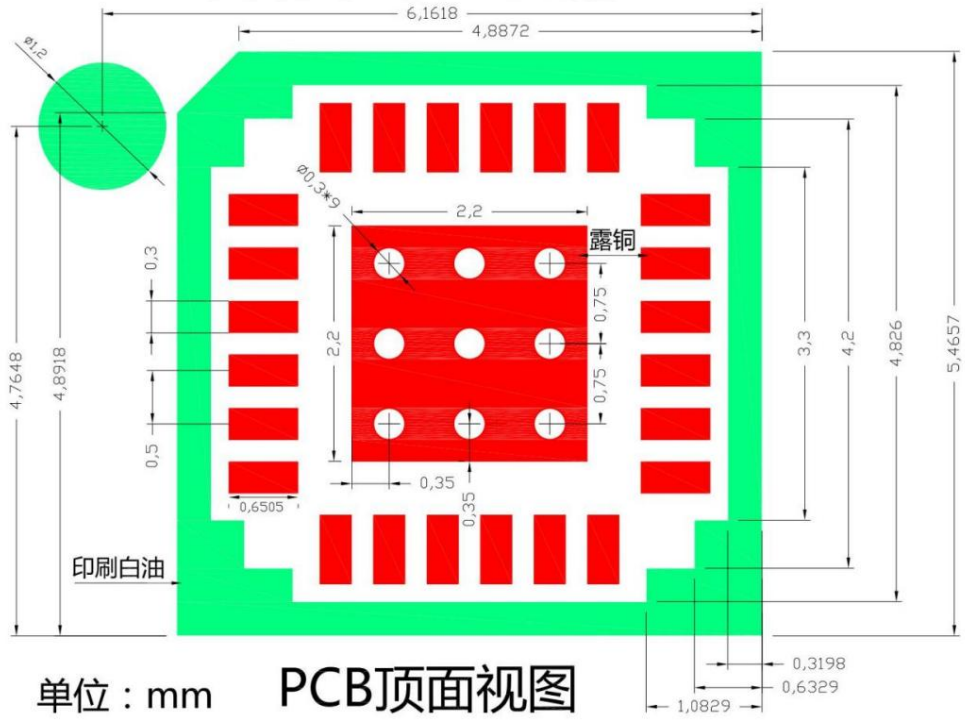


Size / Symbol	Min	TYPE	Max	Size / Symbol	Min	TYPE	Max
A	0.70	0.75	0.80	D1	2.20	2.40	2.55
A1	0.00		0.05	E1	2.20	2.40	2.55
A3	0.203REF			[illegible]	0.50TYP		
b	0.20	0.25	0.30	K	0.20		
D	3.90	4.00	4.10	L	0.30	0.40	0.50
[illegible]	3.90	4.00	4.10				

Note: The above is the chip size, it is recommended that the pin length of the QFN24 library file in the PCB is longer than the chip length, which is convenient for soldering.

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QFN24 PCB Reference



Version history

date	In the first edition	Version
2015.5.20	of the version	Rev1.0
2015.9.16	description , some of the revised indicators are inconsistent; add pin numbers in the circuit diagram. Rev1.1	
2016.2.26	Chip upgrade, simplified peripheral configuration and optimization of	Rev2.0
2016.7.20	grounding scheme Added TP5602B, LED long-light scheme during boost	Rev2.1